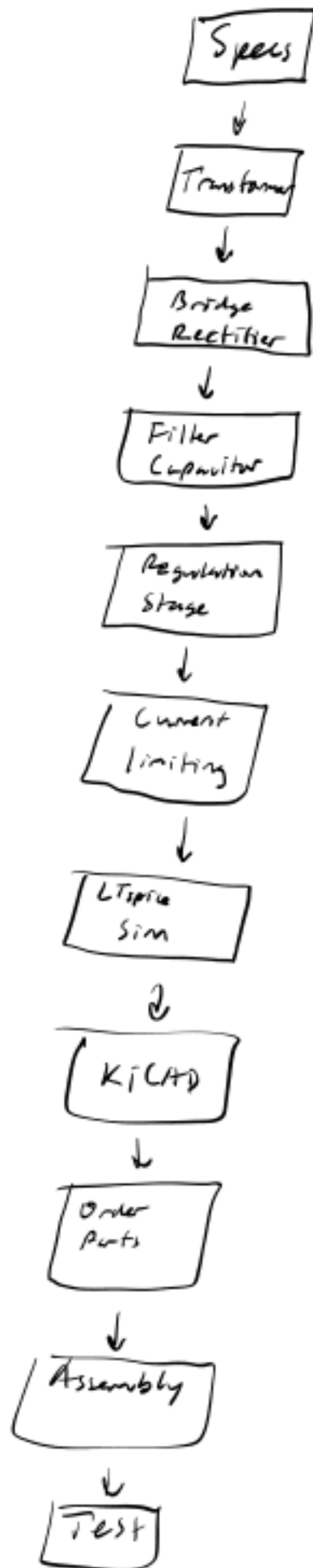


Flow Chart



Output Voltage: 0-15V

Max Current: 1.2 A

Ripple Voltage: $< 50\text{mV}$

Current Limiting: 0.1 - 1 A

PCB: 2 layer, KiCAD

Front Panel: banana jacks, voltage/current knobs, power switch, fuse, status LED

Trade off reasoning (1A instead of 2A)
needs beefier heatsink

Topology Design:

1) Transformer

- Needs headroom above 15V

- RMS voltage $= \frac{(V_{\text{out-max}} + \text{dropout} + \text{ripple allowance})}{1.414}$

- 15V or 18V RMS, rated 1-2A

2) Bridge Rectifier

- full-wave bridge

- bridge above 1A

3) Filter Capacitor Bank

$$C = \frac{I}{2f \cdot V_{\text{ripple-allowed}}}$$

Selecting a Transformer

min V between
in and out
to still
work

1) $V_{\text{min-required}} = V_{\text{out-max}} + V_{\text{dropout}}$

$$V_{\text{min-required}} = 15\text{V} + 2.5\text{V} = 17.5\text{V}$$

2) $V_{\text{peak-required}} = V_{\text{min-required}} + V_{\text{ripple-margin}}$

$$V_{\text{peak-required}} = 17.5\text{V} + 4\text{V} = 21.5\text{V}$$

3) $V_{\text{peak}} = (V_{\text{RMS}} \cdot \sqrt{2}) - V_{\text{diode}}$ $\rightarrow 1.4\text{V} \rightarrow 0.7$ for each bridge

$$21.5 = (V_{\text{RMS}} \cdot \sqrt{2}) - 1.4$$

$$V_{\text{RMS}} = 16.2\text{V}$$

Round up to 18V $\rightarrow$ RMS

common between 15 and 18V $\rightarrow$ more head room

$$4) V_{\text{peak}} = (18V \cdot \sqrt{2}) - 1.4$$

$$V_{\text{peak}} = 24.1V$$

$$5) I_{\text{secondary_rating}} \geq 1.5 \cdot I_{\text{out_max}} = 1.5 \cdot 1.2 = 1.8A$$

NEED TRANSFORMER
18V RMS secondary
 $\geq 1.8A$ rating

Transformer: 266536

Selecting a Bridge Rectifier

$$1) V_{\text{peak}} = V_{\text{RMS}} \cdot \sqrt{2} \\ = 18 \cdot \sqrt{2} \\ = 25.5V$$

$$2) \text{Peak Inverse Voltage (PIV)}$$

$$PIV_{\text{required}} = V_{\text{peak}} \cdot \text{safe_margin}$$

$$PIV_{\text{required}} = 25.5V \cdot 2 \\ = 51V$$

$\times 2 \rightarrow$ industry standard

$$3) I_{\text{rectifier_required}} = I_{\text{load}} \cdot \text{safe_margin}$$

$$I_{\text{rectifier_required}} = 1.2A \times 2 \\ = 2.4A$$

$$4) V_{\text{diode}} = 0.7 \times \underset{2 \text{ diodes}}{2} = 1.4V$$

NEED BRIDGE RECTIFIER
 $PIV \geq 100V$ (rounding up)
Current Rating $\geq 2.4A$

Bridge Rectifier: KBP307

Selecting Filter Capacitors

1) $C = \frac{I_{ld}}{2f \cdot V_{ripple}}$

$\uparrow$ 1A $\downarrow$ 120Hz $\rightarrow$ 4V $\rightarrow$ V-ripple margin

AC $\rightarrow$ 60Hz
with rectifier
one of the negative
humps turn
positive

$\sim \rightarrow \sim$ output are positive
humps which are
now 2
 $2 \times 60\text{Hz} = 120\text{Hz}$

$$C = \frac{1}{2 \cdot 120 \cdot 4} = 1042 \mu\text{F}$$

NEED FILTER
CAPACITOR

$$C \geq 1042 \mu\text{F}$$

Filter
Capacitor: 35VXJ2200M16X25

Regulation Stage

- TL431 (adjustable voltage reference)
- Op-Amp (error amplifier)
- Discrete Pass Transistor (BJT) carrying the load current

1) TL431 Calculation

$V_{out} = V_{ref} \cdot \left(1 + \frac{R_1}{R_2}\right)$ $\rightarrow$ potentiometer

$V_{max} \leftarrow 15\text{V} = 2.5\text{V} \cdot \left(1 + \frac{R_1}{R_2}\right)$ $\rightarrow$ TL431 internal reference $\checkmark$

$$\frac{R_1}{R_2} = 5$$

$$R_1 = 5R_2$$

$$\text{Ex: } R_2 = 2.2\text{k}\Omega$$

$$R_1 = 10\text{k}\Omega \text{ potentiometer}$$

- Runs into topology issue where the range is from 2.5V to 15V

2) $V_{range} = 0\text{V to } 2.5\text{V}$

$$V_{out} = V_{range} \times \text{Gain}$$

$$15 = 2.5 \times \text{Gain}$$

$$\text{Gain} = 6$$

$$3) \text{Gain} = 1 + \frac{R_f}{R_g}$$

$$5 = \frac{R_f}{R_g}$$

$$R_g = 2 \text{ k}\Omega \rightarrow \text{standard value for } R_g$$

$$R_f = 10 \text{ k}\Omega$$

$$\text{Potentiometer (front panel knob)} = 10 \text{ k}\Omega$$

Pass Transistor Stage

- Peak DC after rectification/filtering: $23.3 \text{ V} \rightarrow V_{\text{peak}} = (18 \text{ V} \cdot \sqrt{2}) - 1.1 \text{ V}(2)$
- Min output voltage: 0 V
- Max load current: 1 A

Worst case voltage across transistor

$$\begin{aligned} V_{\text{transistor}} &= V_{\text{in}} - V_{\text{out}} \\ &= 23.3 \text{ V} - 0 \text{ V} \\ &= 23.3 \text{ V} \end{aligned}$$

Worst case power dissipation

$$\begin{aligned} P_{\text{transistor}} &= V_{\text{transistor}} \times I_{\text{load}} \\ &= 23.3 \text{ V} \times 1.2 \text{ A} \\ &= 27.96 \text{ W} \end{aligned}$$

Needs:

collector-emitter voltage: $\geq 30 \text{ V}$

collector current: $\geq 2 \text{ A}$

Power Dissipation: $\geq 28 \text{ W}$ with heatsink

Heat Sink Calculation

1) $P = 27.96W$

2) $T_j = \text{junction temperature } (\leq 125^\circ C)$

$T_{\text{ambient}} = \text{surrounding air temp (worst case } 40^\circ C \text{ indoor electronics)}$

$R_{\theta JC} = \text{junction-to-case thermal resistance } (1-2^\circ C/W)$

$R_{\theta CS} = \text{case-to-sink thermal resistance } (0.5-1^\circ C/W)$

$R_{\theta SA} = \text{sink-to-ambient thermal resistance}$

3) $R_{\theta SA} = (T_j - T_{\text{ambient}}) / (P - R_{\theta JC} - R_{\theta CS})$

$$= (100^\circ - 40^\circ) / 27.96W - 1.92^\circ C/W - 0.75^\circ C/W$$
$$= 85 / 27.96 - 2.25$$
$$= 0.37^\circ C/W$$

Parts chosen: 2SC2837

Heatsink: Wakefield 421K

Fan: Winsing 40mm Fan

Current Limiting Stage

Using the specs of TIP41C

$h_{FE} = 50$ (min)
 $\downarrow$
DC current gain

$\odot I_C = 3A, V_{CE} = 4V$

$V_{CE(sat)} = 2V_{max} \odot I_C = 5A, I_B = 0.5A$

Current sense resistor:

ALD3000005007J4L00

Current limiting transistor:

2N3904TA

1) $I_B = I_{I_{dmax}} / h_{FE_{min}}$

$$= 1.2A / 50$$
$$= 0.024A$$

2) $R_{sense} = V_{BE} / I_{I_{dmax}}$

$$= 0.6V / 1.2A$$
$$= 0.5\Omega$$

Standard small transistor

$P_{sense} = I^2 \times R_{sense}$

$$= (1.2)^2 \times 0.5\Omega$$
$$= 0.72W$$

Protection Circuits

1) Reverse Polarity Calculation:

$$I_{\text{diode-required}} = 2 \times 1.2A = 2.4A$$

$$V_{\text{reverse-required}} \geq 15V$$

Target Spec: Schottky diode $\geq 2.4A$, $\geq 20-30V$ reverse rating

Diode: 1N5822

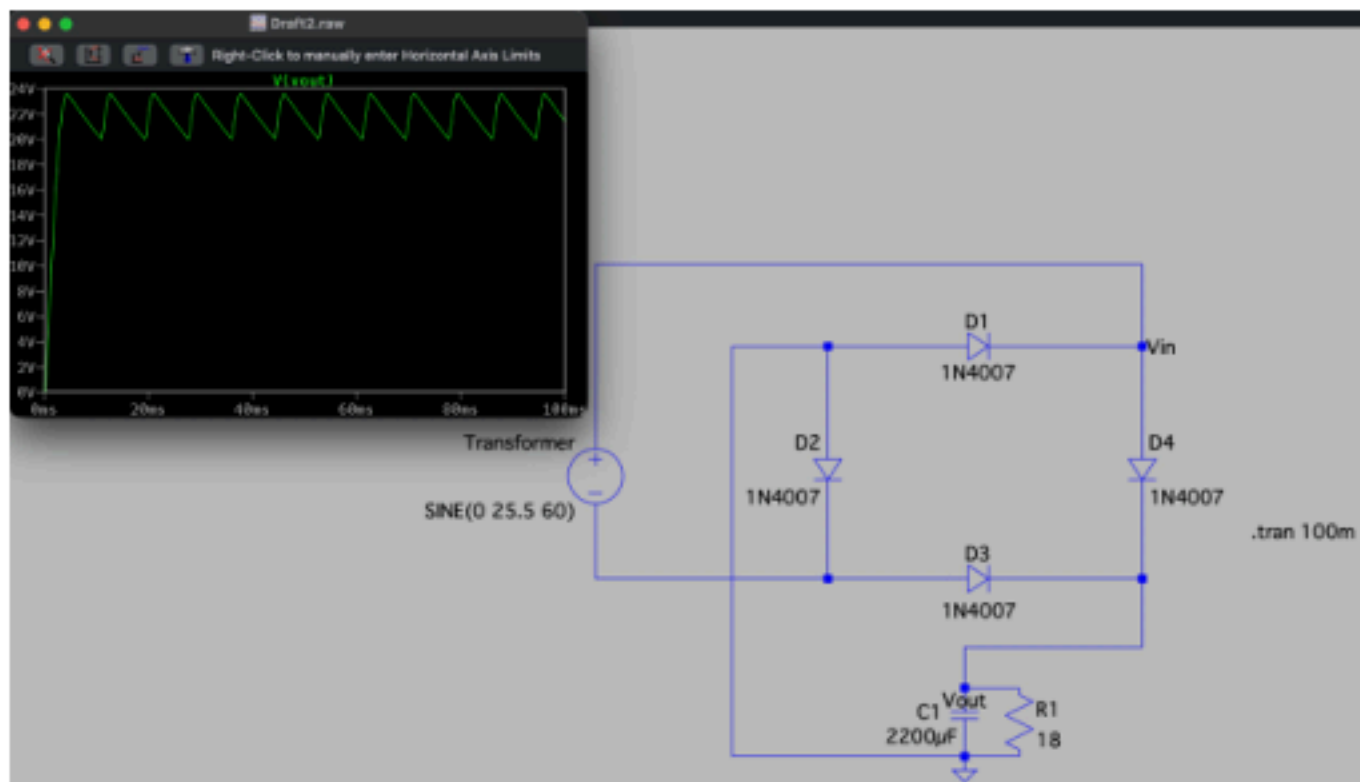
Relay: G6RN-1DC5

2) Relay sizing:

$$\text{Contact current rating} \geq 2 \times I_{\text{max}} = 2.4A$$

Coil voltage 5V or 12V

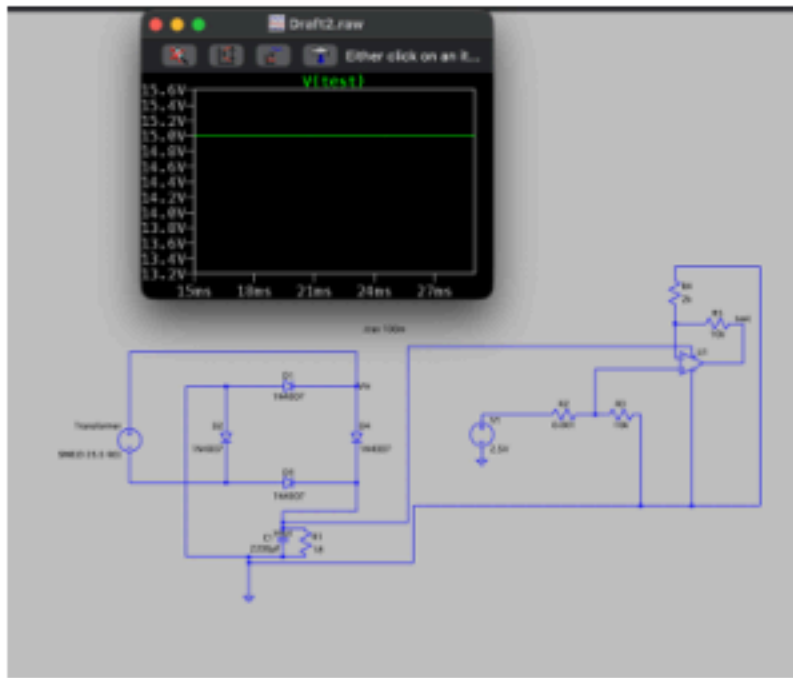
Transformer Circuit (Bridge Rectifier)



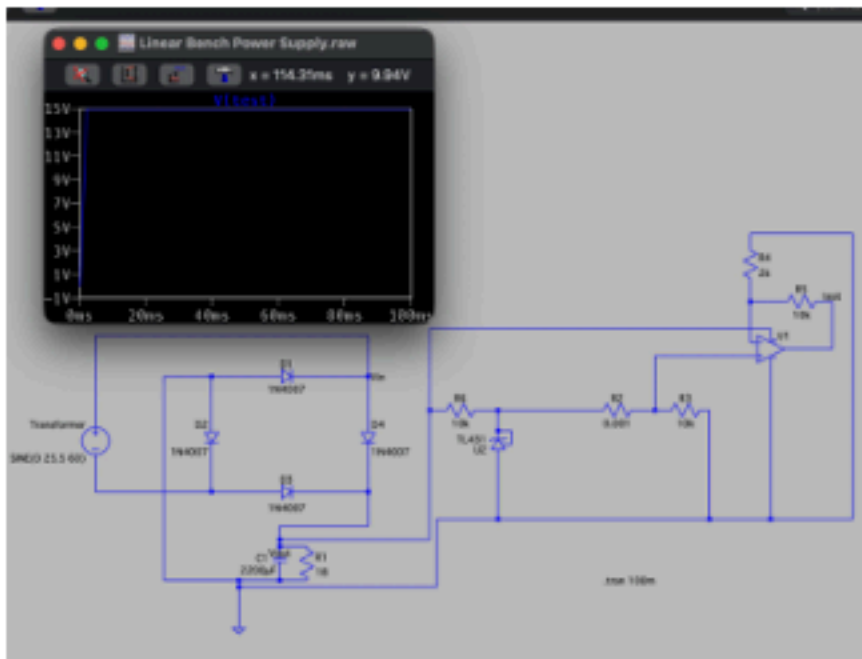
V_{min} in this circuit was 20.03V

V_{min} we calculate was 17.5V

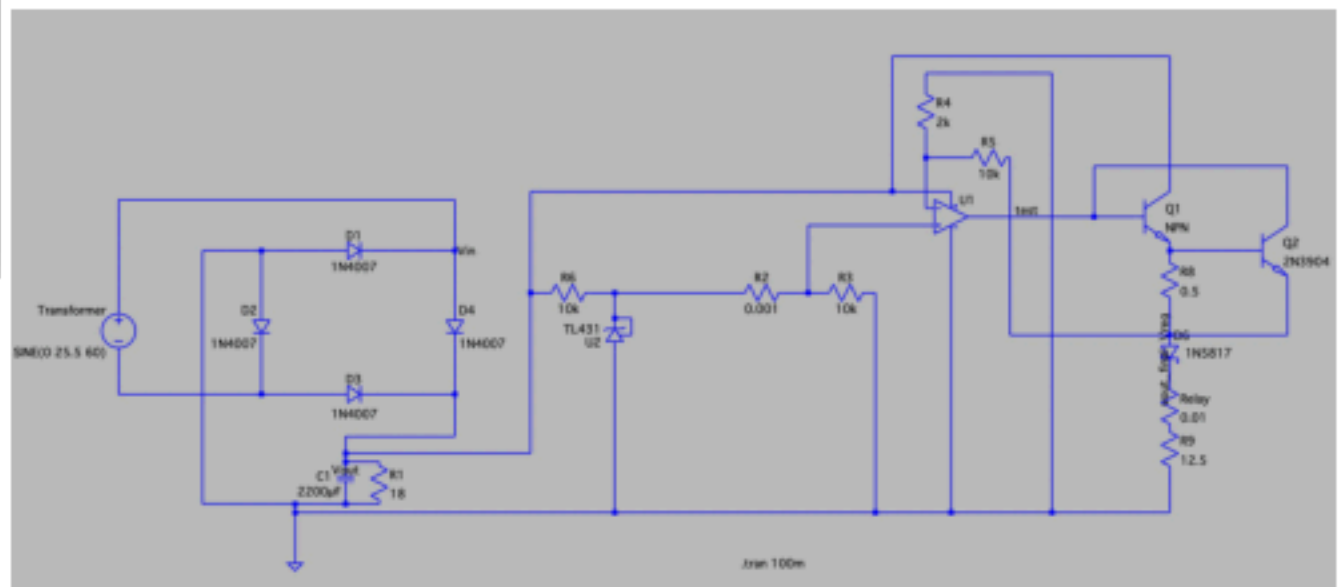
This is still good since we calculated for the lowest voltage to be 17.5V.



- Needed stable 2.5V
- Used 2.5V voltage source sub.
- built the regulation stage

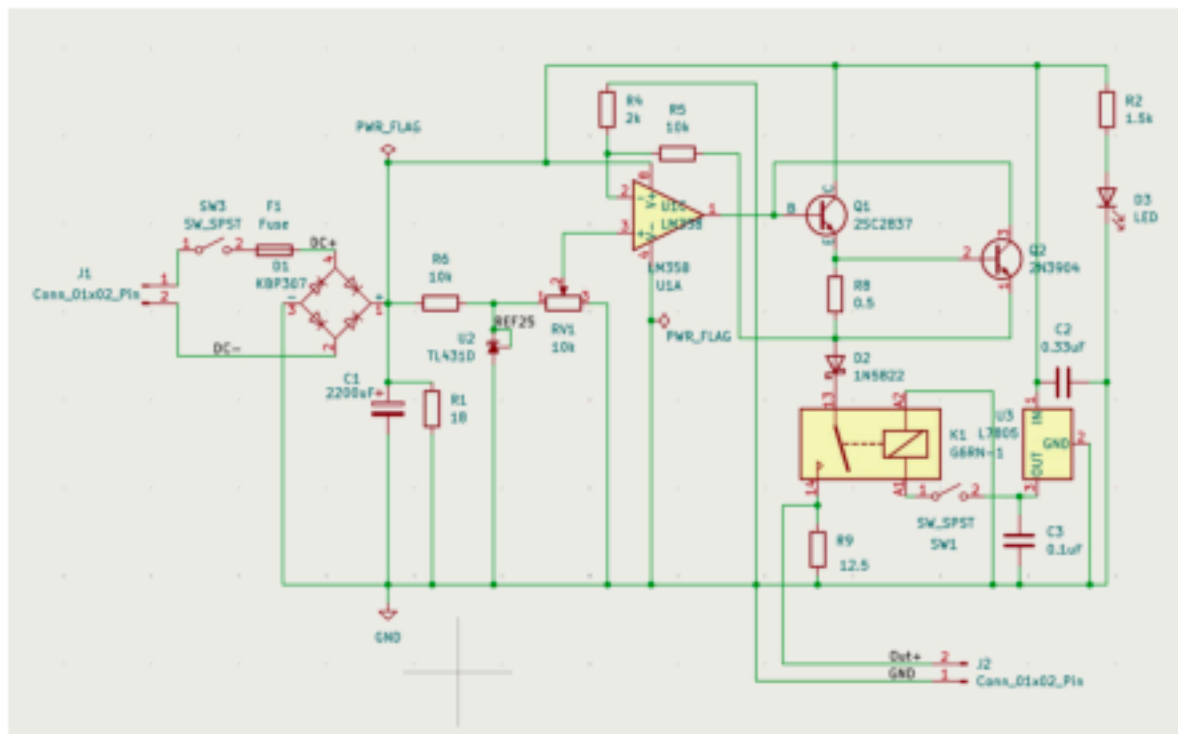


- Remove 2.5 voltage source
- TL431 implementations

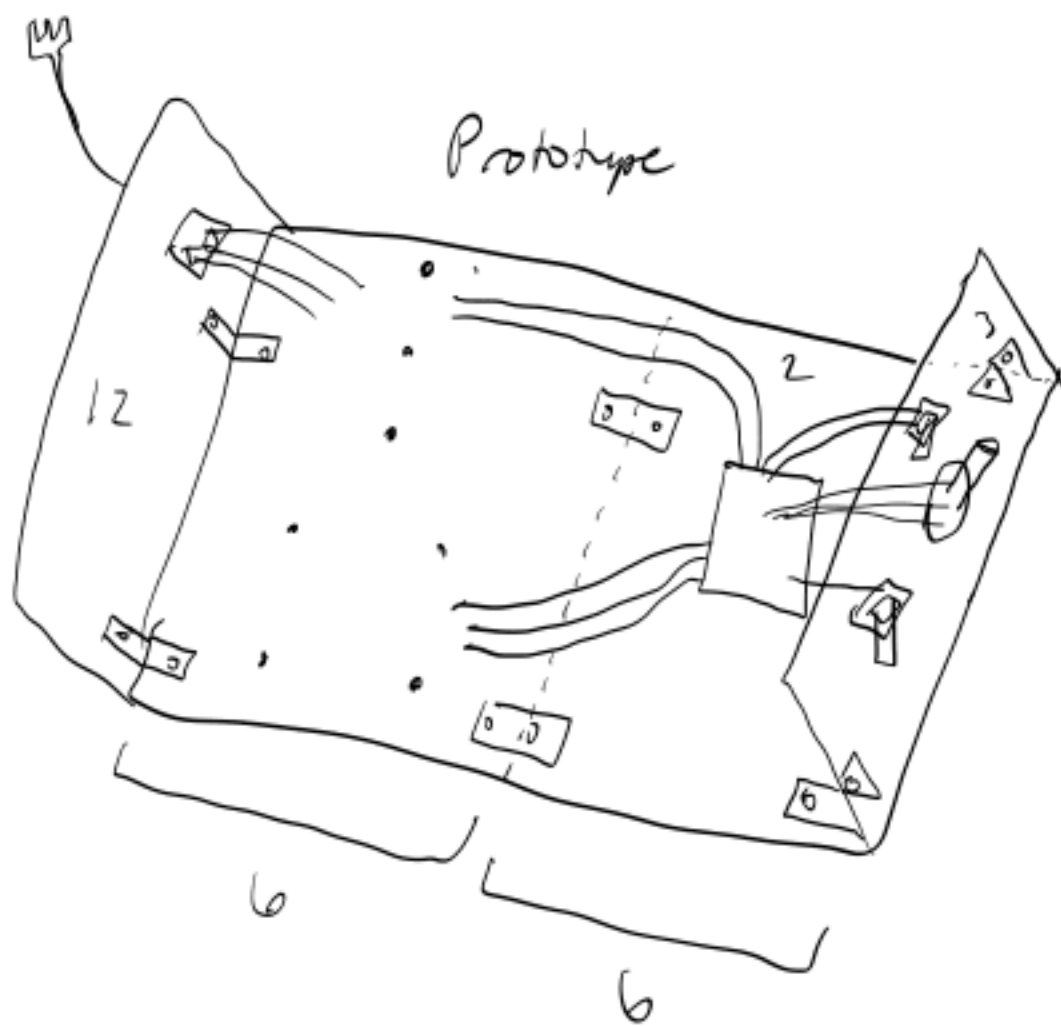
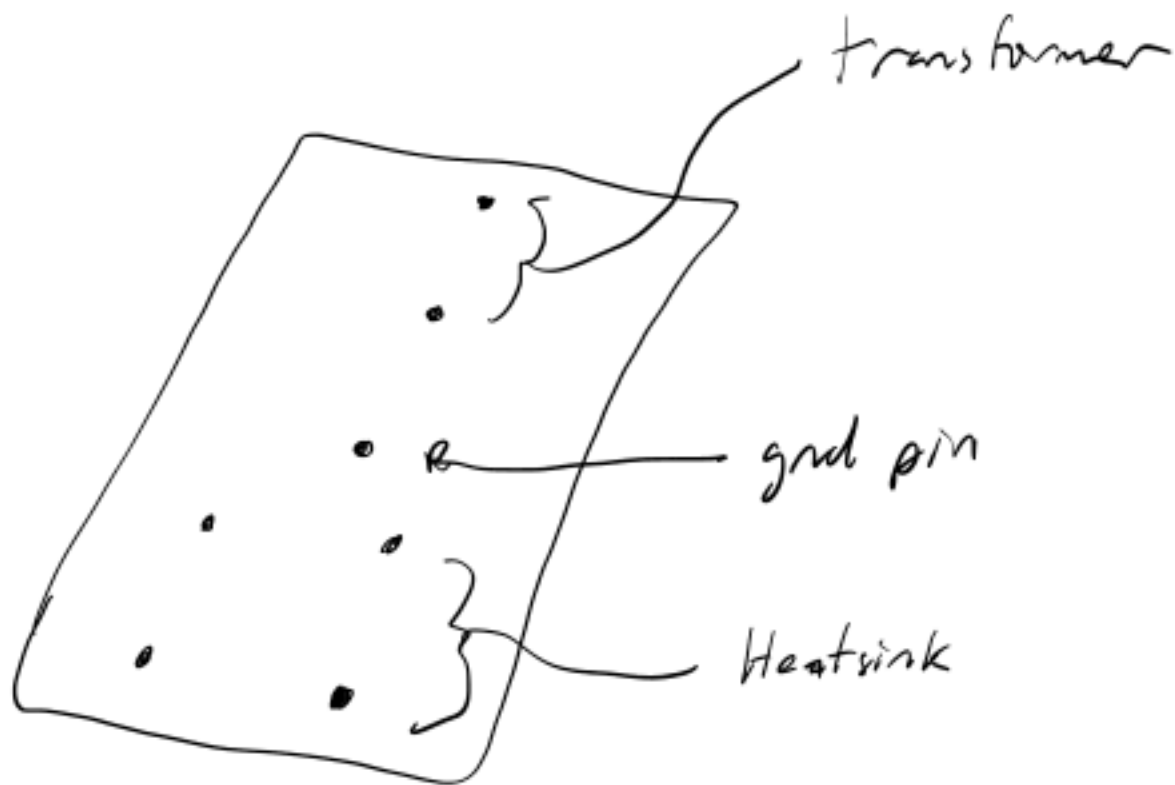


Final Schematic of Linear Bench Power Supply

- Added both transistors and tested output to match calculations
- Added Protection circuit as well in case of incorrect wiring
 - current limiting in case of excessive current draw
 - relay for safety use



Final Kicad Schematic



12 x 12